Urban Heat Island and Equity Analysis in Boston, MA

Using Landsat 9 and U.S. Census Data

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4 May 2026

Study Area and Data

This study examines urban heat island patterns across Boston, Massachusetts and adjacent municipalities using census tracts as the unit of analysis. The satellite imagery used in the study was collected on July 30, 2025, which was selected based on minimal cloud cover and high air temperatures (~86°F), to provide reliable land surface temperature data and to capture peak urban heat island conditions.

Datasets: Landsat 9 (Collection 2 Level-2), 2020 Census Tracts (MassGIS), 2022 American Community Survey 5-year estimates; accessed via NHGIS (IPUMS).

Methods

Satellite imagery and demographic data were processed and integrated using QGIS and Google Sheets to analyze spatial patterns in urban heat and social vulnerability.

Raster Processing

Landsat 9 imagery was used to derive land surface temperature (LST), vegetation, and built-up indices. All raster datasets were reprojected to a common coordinate system (EPSG:32619 [WGS84 / UTM Zone 19N]) and clipped to the study area. Land Surface Temperature (LST) was calculated from the Landsat thermal band and converted from Kelvin to Celsius. Normalized Difference Vegetation Index (NDVI) was calculated using the red and near infrared bands:

$$NDVI = (NIR - Red) / (NIR + Red)$$

Normalized Difference Built-up Index (NDBI) was calculated using shortwave infrared and near-infrared bands:

$$NDBI = (SWIR - NIR) / (SWIR + NIR)$$

These variables (LST, NDVI, NDBI) were aggregated to the census tract level using zonal statistics to calculate mean values for each tract. These values were the basis for statistical and spatial analyses.

Demographic Data Integration

Demographic variables from the 2022 American Community Survey were joined to census tract geometries using a common geographic identifier. Derived variables included:

$$\text{Poverty Rate}_{(pov\_rate)} = \text{population below poverty}_{(AQ5ZE002)} / \text{total population}_{(AQ5ZE001)}$$

$$\text{Percent Minority}_{(pct\_minority)} = (\text{total population}_{(AQ5ZE001)} - \text{white population}_{(AQNOE003)}) / \text{total population}_{(AQ5ZE001)}$$

Statistical Analysis

A correlation analysis was performed in Google Sheets to examine relationships between LST and independent variables (NDVI, NDBI, poverty rate, percent minority, and median household income). A linear regression model between LST and NDVI was used to predict expected surface temperatures:

$$LST_{predicted} = -34.2 * NDVI + 46.8$$

Residual LST was calculated as the difference between observed and predicted LST to identify tracts where temperatures are higher or lower than expected based on vegetation alone.

Heat Vulnerability Index (HVI)

A Heat Vulnerability Index (HVI) was created by normalizing and averaging LST, percent minority, and poverty rate variables. Census tracts in the top 20% of HVI values were identified as highly vulnerable populations.

$$HVI = (LST_{norm} + pov\_rate_{norm} + pct\_minority_{norm}) / 3$$

Priority Heat Intervention Area Identification

Priority heat intervention areas were identified by combining binary values for the top 10% highest LST, 25% highest NDBI, and 20% highest HVI tracts. A priority score was calculated as the sum of these values:

$$Priority = LST_{bin} + NDBI_{bin} + HVI_{bin}$$

Tracts with a score of 3 are areas where high environmental (LST and NDBI) and social (HVI) risk factors overlap.

Results

Urban Heat Distribution

Land surface temperatures are highest in Roxbury, Dorchester, South Boston, Allston, Cambridge, Somerville, Everett, Medford, Chelsea, and East Boston. Cooler areas are in coastal zones and higher vegetation cover in Winthrop, Brookline, Milton, and Quincy.

Variables	Correlation (r)	Relationship
LST vs NDVI	-0.59	Strong negative correlation
LST vs NDBI	0.62	Strong positive correlation
LST vs Poverty Rate	0.12	Very weak positive correlation
LST vs % Minority	0.26	Weak positive correlation
LST vs Median Income	-0.03	Almost no correlation

Figure 1. Correlation between land surface temperature and independent variables.

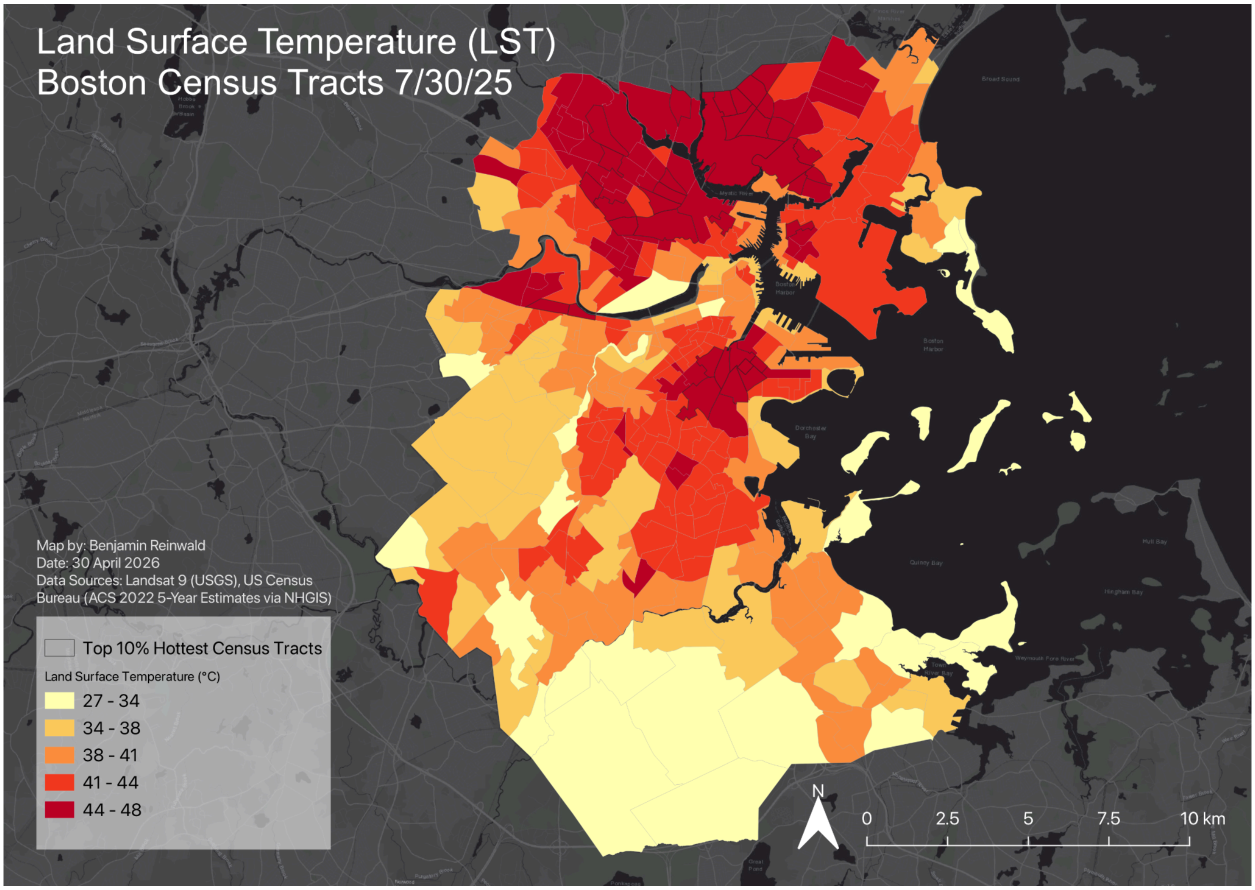

Figure 2. Map showing land surface temperature distribution across Boston, MA on July 30, 2025.

Vegetation and Built Environment Effects

The regression model (y=-34.2x+46.8) indicates an inverse relationship between NDVI and land surface temperature. A 0.1 increase in NDVI corresponds to an approximate 3.4°C decrease in surface temperature. The model explains approximately 35% of the variation in LST ($R^2 = 0.35$).

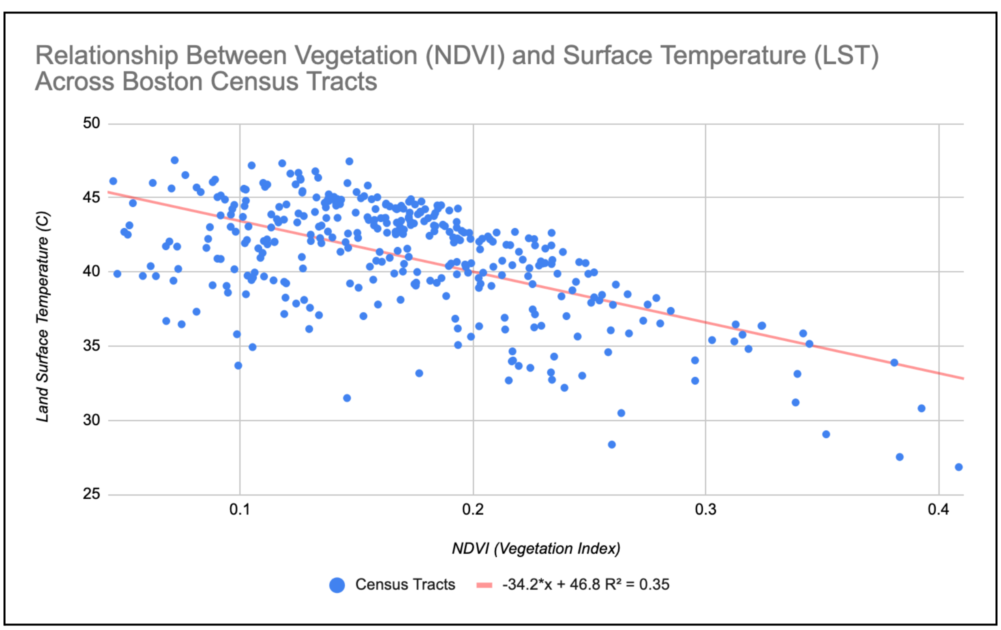

Figure 3. Scatter plot showing the relationship between vegetation and land surface temperature across Boston census tracts (July 30, 2025).

The regression model (y=65.9x+45.4) indicates a positive relationship between built-up intensity (NDBI) and land surface temperature. A 0.1 increase in NDBI corresponds to an approximate 6.6°C increase in surface temperature. The model explains approximately 38% of the variation in LST ($R^2 = 0.38$).

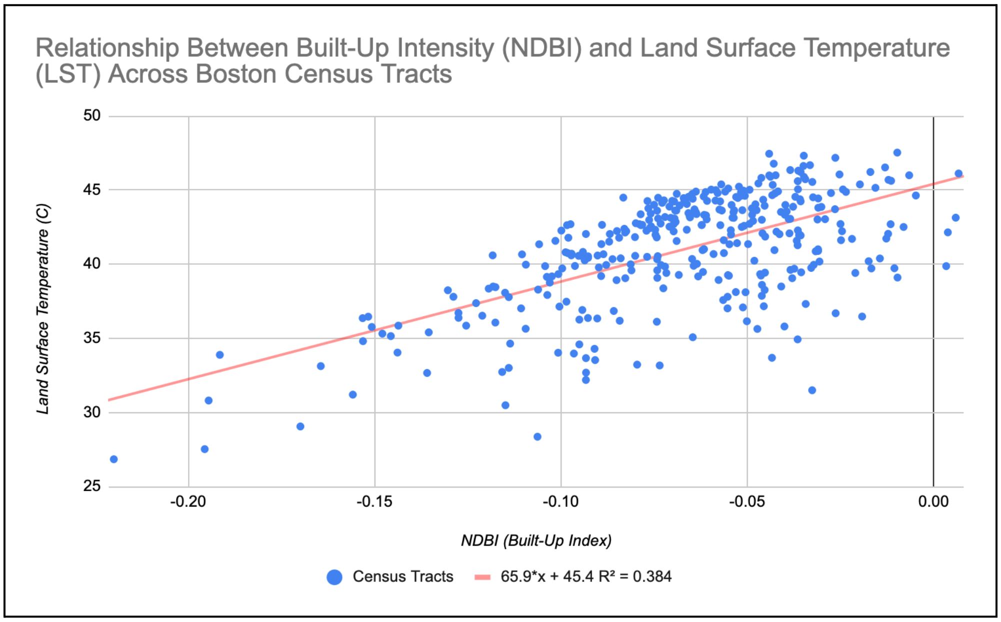

Figure 4. Scatter plot showing the relationship between built-up intensity and land surface temperature across Boston census tracts (July 30, 2025).

Residual Heat

The residual land surface temperature map shows where temperatures are higher or lower than expected based on vegetation alone. Several neighborhoods in northern Boston and parts of Roxbury, Dorchester, South Boston, and East Boston exhibit positive residuals, suggesting factors beyond vegetation, such as building density and impervious surfaces, are contributing to higher temperatures. Coastal areas exhibit negative residuals likely due to cooling effects of the Atlantic Ocean and the Charles River.

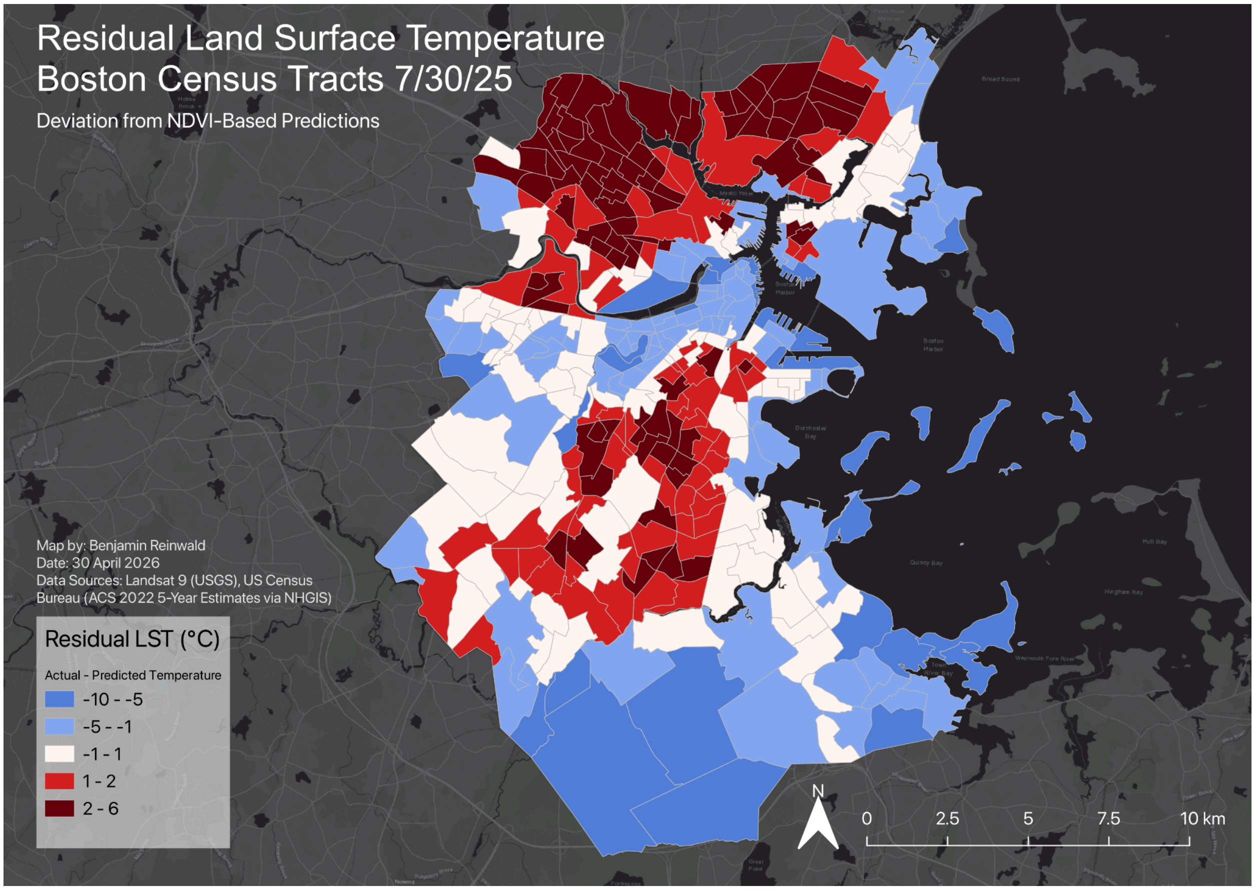

Figure 5. Map showing areas LST is hotter or cooler than predicted by the regression model (y = -34.2x + 46.8) where y = LST and x = NDVI.

Social Vulnerability and Heat

Areas with higher minority populations show slightly raised LST values (r = 0.26), while poverty rate shows a weaker relationship with LST (r = 0.12). This suggests that social factors alone do not explain temperature variation, but there is some spatial overlap between vulnerable populations and higher heat exposure.

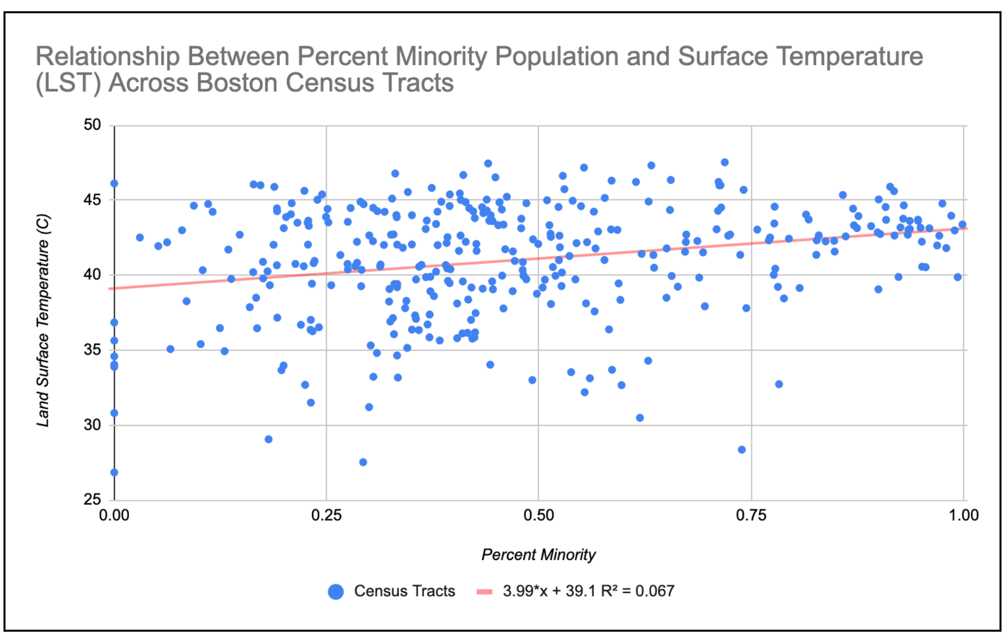

Figure 6. Scatter plot showing the relationship between land surface temperature and percent minority population across Boston census tracts (July 30, 2025).

The regression model (y=3.99x+39.1) indicates a weak positive relationship between percent minority population and land surface temperature. A 0.1 increase in percent minority is associated with an approximate 0.39°C increase in surface temperature. The model explains approximately 6.7% of the variation in LST ($R^2 = 0.067$).

Heat Vulnerability Index

The Heat Vulnerability Index (HVI) combines LST, percent minority, and poverty rate to highlight areas of heightened vulnerability. Census tracts in Roxbury, the South End, South Boston, Dorchester, Allston, Medford, Everett, Revere, and Chelsea contain the highest HVI values. These are locations where both environmental and social risk factors are present.

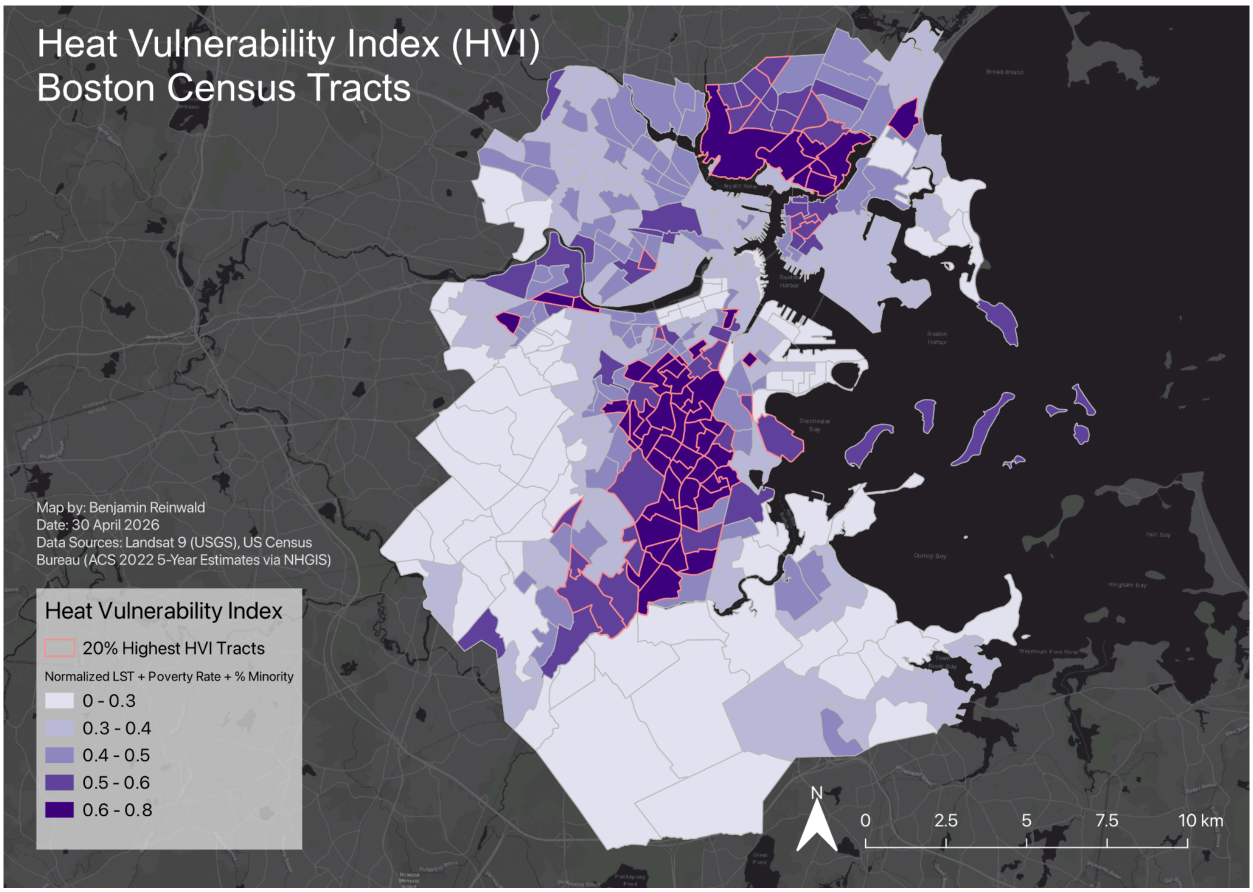

Figure 7. Map showing the Heat Vulnerability Index across Boston census tracts. The 20% highest HVI tracts are outlined in red.

Quantitative Comparison

Census tracts in the top 20% of heat vulnerability experience higher average land surface temperatures compared to the rest of the study area.

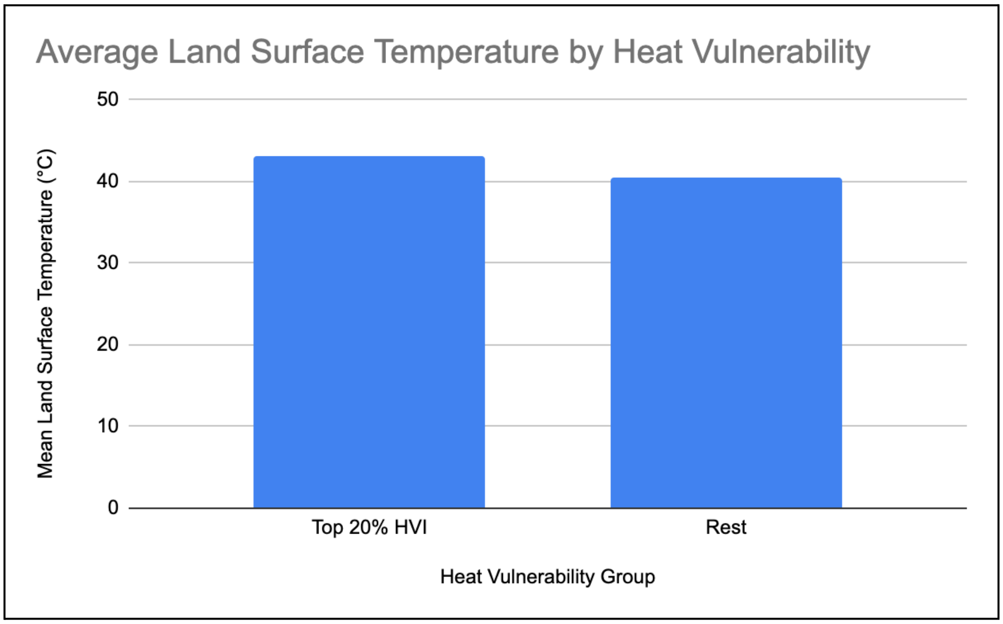

Figure 9. Bar chart comparing the average LST between the top 20% highest HVI tracts and the rest of the study area.

There is a wider temperature range observed in tracts with a lower percentage of minority population, likely representing the presence of both dense urban areas and neighborhoods with more vegetation.

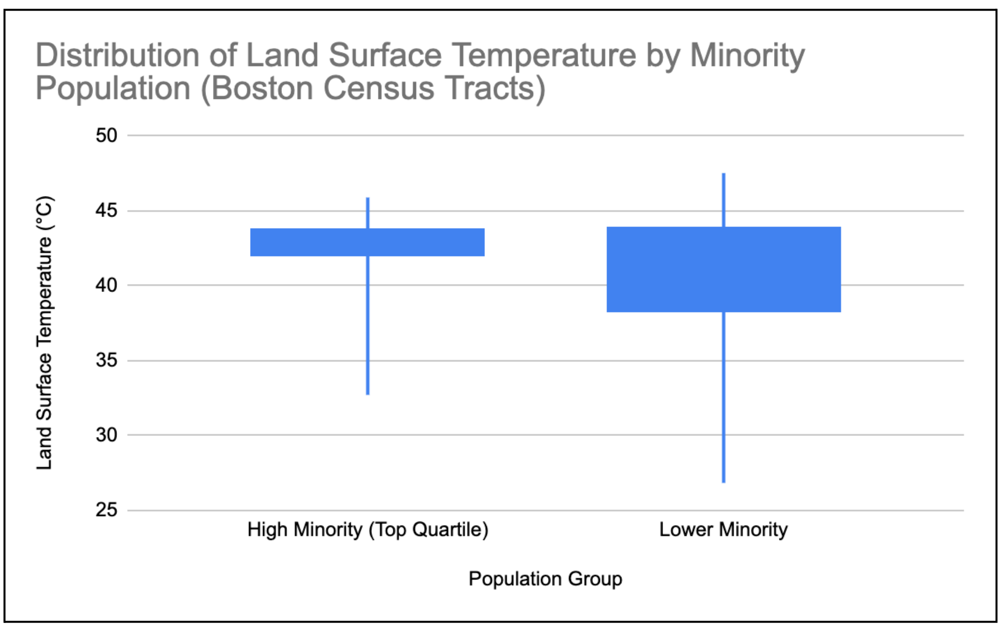

Figure 10. Boxplot showing the distribution of LST by minority population group.

Urban Heat Intervention Priority Areas

Priority intervention areas were identified by combining the 10% hottest (LST), 25% most built-up (NDBI), and 20% highest vulnerability (HVI) tracts. Areas designated as high priority are tracts where all three risk factors overlap. Moderate priority tracts have two overlapping risk factors and low priority tracts have one. The highest priority areas are in South Boston, the South End, Roxbury, Allston, Chelsea, Everett, Medford, and East Boston.

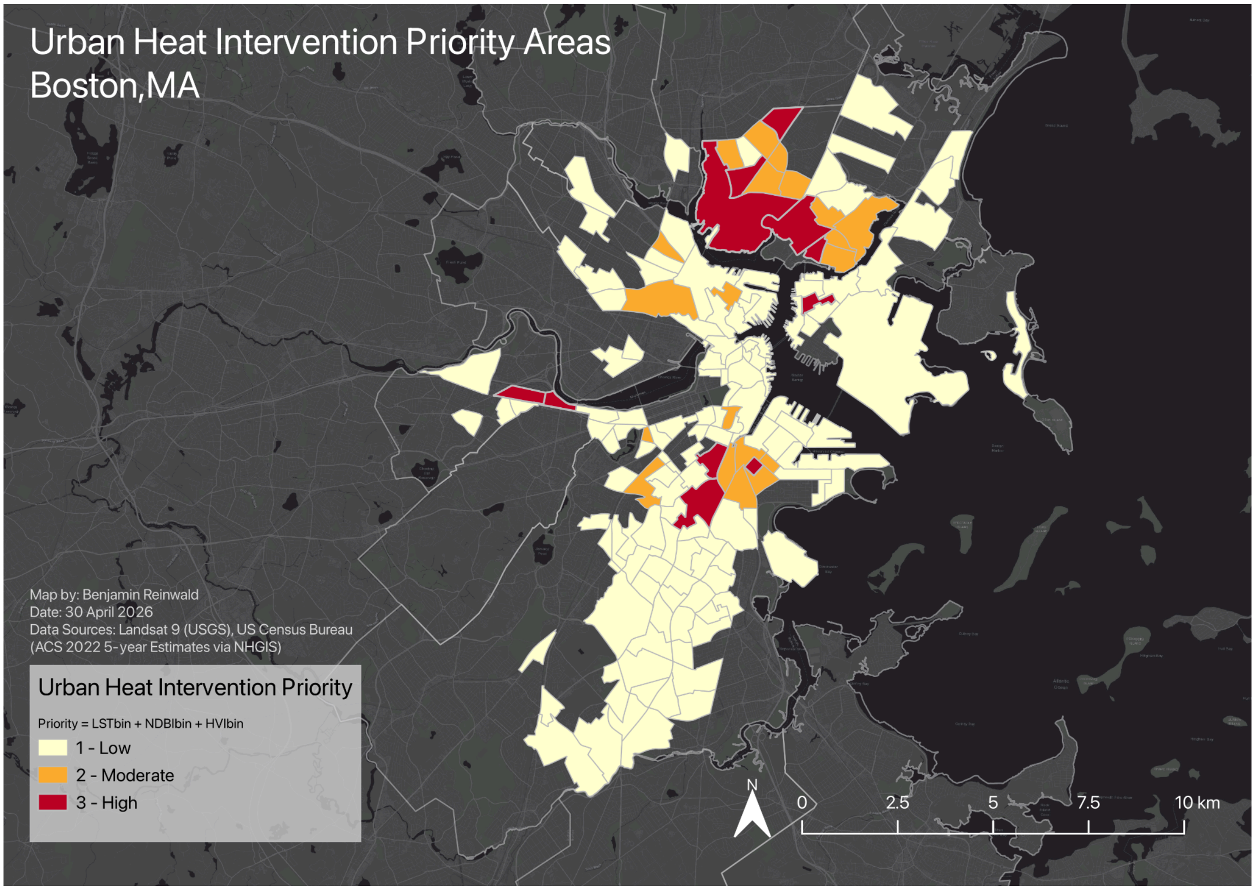

Figure 8. Map showing priority areas for urban heat intervention in Boston, MA.

Conclusion

Urban heat island patterns in Boston and adjacent municipalities are primarily driven by the built environment, and socially vulnerable populations are located disproportionately in areas with higher exposure to extreme heat. Vegetation and built-up intensity are the strongest drivers of temperature variation. Social variables such as minority population and poverty level show weaker but still meaningful spatial relationships with higher land surface temperature.

Residual analysis identified census tracts that are hotter or cooler than expected based on vegetation alone, indicating that variables like building density, impervious surfaces, and proximity to the Atlantic Ocean have a significant effect on land surface temperature distribution.

By combining environmental and social factors, priority urban heat intervention areas were identified in parts of South Boston, the South End, Roxbury, Allston, Chelsea, Everett, Medford, and East Boston. These areas would benefit from targeted urban cooling strategies such as green infrastructure, reflective pavements or roofing, tree planting, and water features. Increasing vegetation in these areas is likely to help reduce surface temperatures based on the observed relationship between vegetation and land surface temperature.